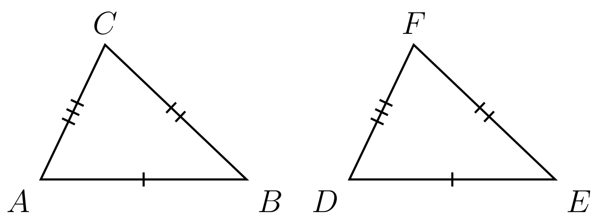
Reading the Evidence for Triangle Congruence

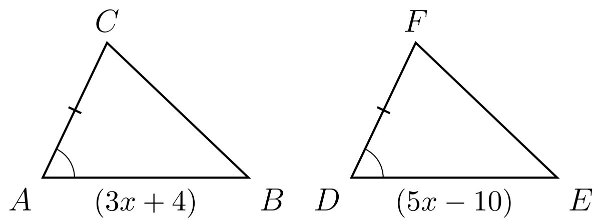
- Matching tick marks mean congruent *sides*; matching arcs mean congruent *angles*. Read the marks before you name anything.
- The five criteria are **SSS**, **SAS** (the angle must be *between* the two sides), **ASA** (the side must be *between* the two angles), **AAS** (the side is *not* between them), and **HL** (right triangles only: hypotenuse and one leg).
- Side-side-angle is not on the list. Two triangles can match that way and still differ.

1. The tick marks show which sides are congruent.



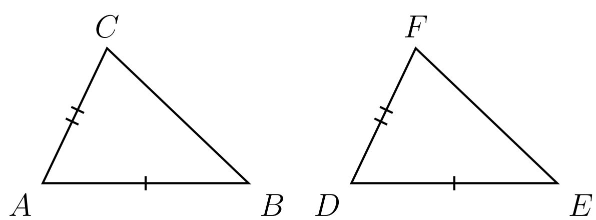
Name the congruence criterion these marks establish, and write the three congruence statements it rests on.

2. $\triangle ABC \cong \triangle DEF$ by SAS: the marked sides and the marked included angles correspond. In centimetres, $AB = 3x + 4$ and $DE = 5x - 10$. Find x.



Answer: ____________________

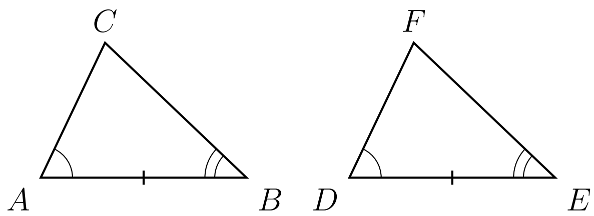
3. Only two pairs of sides are marked congruent.



(a) Name one further piece of evidence that would complete an SAS argument, and say exactly where it must sit.

(b) Name a different piece of evidence that would instead complete an SSS argument.

4. $\triangle ABC \cong \triangle DEF$ by ASA: the marked side lies between the two marked angles. In degrees, $m\angle A = 4x + 6$ and $m\angle D = 6x - 14$.



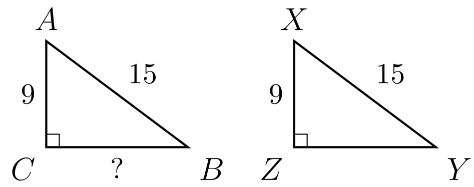
(a) Find x.

Answer: ________________

(b) Find $m\angle A$.

Answer: ________________ degrees

5. $\triangle ABC$ and $\triangle XYZ$ are right triangles with their right angles at C and at Z. Lengths are in centimetres.

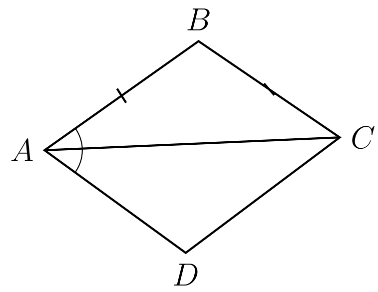


(a) Name the criterion that proves these two triangles congruent, and say why the ordinary side-side-angle pattern is not enough for triangles that are not right triangles.

(b) Find BC, the unmarked leg of $\triangle ABC$.

Answer: ________________ cm

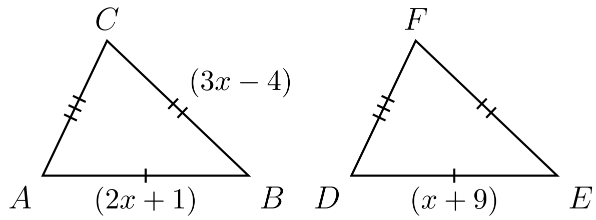
6. In quadrilateral $ABCD$ the diagonal $\overline{AC}$ has been drawn, splitting it into $\triangle ABC$ and $\triangle ADC$. The marks show $\overline{AB} \cong \overline{AD}$ and $\angle BAC \cong \angle DAC$.



(a) The two triangles share side $\overline{AC}$. Name the property that lets you write $\overline{AC} \cong \overline{AC}$ as a step, and explain why it is a legitimate line in a proof.

(b) With that step added, name the criterion that proves the two triangles congruent.

7. $\triangle ABC \cong \triangle DEF$ by SSS. In centimetres, $AB = 2x+1$, $DE = x+9$, and $BC = 3x-4$.



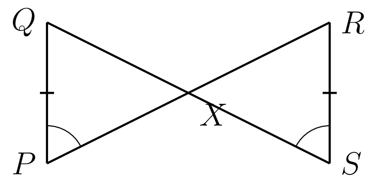
(a) Find x.

Answer: ________________

(b) Find EF, the side of $\triangle DEF$ corresponding to $\overline{BC}$.

Answer: ________________ cm

8. $\overline{PR}$ and $\overline{QS}$ intersect at X. The marks show $\overline{PQ} \cong \overline{SR}$ and $\angle QPX \cong \angle RSX$.



(a) There is a third pair of congruent parts in this figure that no tick mark shows. Name it, name the theorem that supplies it, and then name the criterion that proves $\triangle PQX \cong \triangle SRX$.

(b) A classmate says the same figure would prove the triangles congruent by SSA if $\overline{QX} \cong \overline{RX}$ were marked instead of the angle pair. Explain why that argument fails.